

Overview

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Markov processes

The very simplest type of stochastic process is a **random process**

$$W_2(x_1, t_1, x_2, t_2) = W_1(x_1, t_1)W_2(x_2, t_2) \implies R(\tau) = c\delta(\tau)$$

Next simplest: all information is in W_2 — **Markov process**

Define **transition probabilities**

$$P(x_2 t_2 | x_1 t_1) \equiv T(x_1, t_1 \rightarrow x_2, t_2) \quad \int P(x_2 t_2 | x_1 t_1) dx_2 = 1$$
$$W_2(x_1, t_1, x_2, t_2) = W_1(x_1, t_1)P(x_2 t_2 | x_1 t_1)$$

All information about dynamics of the process is in T .

Where we go next is independent of how we got there.

$$P(x_n t_n | x_{n-1} t_{n-1} \dots x_1 t_1) = P(x_n t_n | x_{n-1} t_{n-1})$$

Only the most recent time counts!

Markov processes

Examples of Markov processes

- random walk
- population dynamics

Examples of non-Markov processes

- self-avoiding random walk
- stock markets

Markov chains

With discrete time steps: **Markov chain**

— need only look at transition probabilities from one time step to next

$$T(Xt_n \rightarrow Yt_{n+1}) \equiv T(X \rightarrow Y)$$

The Master Equation

Look at probability of being in state X at time t , $P(X, t)$

$$P(X, t_{n+1}) = \sum_Y P(Y, t_n) T(Y \rightarrow X)$$

$$P(X, t_n) = \sum_Y P(X, t_n) T(X \rightarrow Y)$$

Master equation

$$P(X, t_{n+1}) - P(X, t_n) = \sum_Y \left[P(Y, t_n) T(Y \rightarrow X) - P(X, t_n) T(X \rightarrow Y) \right]$$

Continuous time:

$$\frac{\partial P(x, t)}{\partial t} = \int \left[P(y, t) w(y, x) - P(x, t) w(x, y) \right] dy$$

Detailed balance

For a **stationary process** $P(X, t_{n+1}) = P(X, t_n) = P(X)$

$\Rightarrow$ rhs of Master Equation is 0.

Sufficient condition

$$P(X)T(X \rightarrow Y) = P(Y)T(Y \rightarrow X) \quad \text{Detailed balance}$$

Can prove that Master Eq describes irreversible process towards equilibrium

Markov chain Monte Carlo

Use such a process to create configurations X
with equilibrium distribution $P(X)$

Recap

Detailed balance

Master equation for a Markov chain:

$$P(X, t_{n+1}) - P(X, t_n) = \sum_Y \left[P(Y, t_n) T(Y \rightarrow X) - P(X, t_n) T(X \rightarrow Y) \right]$$

Detailed balance: $P(X) T(X \rightarrow Y) = P(Y) T(Y \rightarrow X)$

Sufficient, but not necessary condition for a stationary process!

Markov chain Monte Carlo

Task

Generate an ergodic Markov chain of configurations which leads to the stationary distribution $P(X) = \exp(-\beta H(X))$

Main criterion: detailed balance

$$P(X)T(X \rightarrow Y) = P(Y)T(Y \rightarrow X)$$

Trial and acceptance step

Write transition probability as $T(X \rightarrow Y) = \omega_{XY}A_{XY}$

- $\omega_{XY} = \omega_{YX}$ = trial step probability, $\sum_Y \omega_{XY} = 1$.
 - $A_{XY} \neq A_{YX}$ = acceptance probability
- 1 'Random walk' from X to somewhere else
 - 2 Accept this trial step with probability A_{XY} , otherwise stay put

Metropolis algorithm

Use the acceptance probability

$$A_{XY} = \begin{cases} 1 & P(X) \leq P(Y) \\ \frac{P(Y)}{P(X)} & P(X) > P(Y) \end{cases}$$

Proof.

of detailed balance

$$\frac{T(X \rightarrow Y)}{T(Y \rightarrow X)} = \frac{\omega_{XY} A_{XY}}{\omega_{YX} A_{YX}} = \frac{A_{XY}}{A_{YX}}$$

If $P(X) \leq P(Y)$ then $A_{XY} = 1$, $A_{YX} = P(X)/P(Y)$

If $P(X) > P(Y)$ then $A_{XY} = P(Y)/P(X)$, $A_{YX} = 1$

In both cases

$$\frac{T(X \rightarrow Y)}{T(Y \rightarrow X)} = \frac{A_{XY}}{A_{YX}} = \frac{P(Y)}{P(X)}$$

which is just the detailed balance condition!



Metropolis algorithm

Ergodicity

depends on the properties of ω_{XY}

It must be possible to reach any Y from any X after a series of steps.

If you start far from equilibrium, any trial step that brings you closer to equilibrium will be accepted!

Statistical physics

$$P(X) \propto e^{-\beta H(X)} \implies \frac{P(Y)}{P(X)} = e^{-\beta \Delta H}$$

We do not need to know overall normalisation $\frac{1}{Z} = [\sum_X e^{-\beta H(X)}]^{-1}$

This is good, since

- it can be infinite, and
- it is usually impossible to compute

Autocorrelations

Two successive configurations in Markov chain will be “close” in phase space $\implies$ any quantity will have similar values on the two configs

The two are correlated: $R(1) = \langle \Phi[X_n]\Phi[X_{n+1}] \rangle \neq \langle \Phi \rangle^2$

For Monte Carlo we need samples to be statistically independent

Autocorrelation time τ

$$R(t) = \langle \Phi[X_n]\Phi[X_{n+t}] \rangle - \langle \Phi \rangle^2 = Ce^{-t/\tau}$$

Measures the “memory” of the Markov process

Configurations separated by $t \gtrsim 2\tau$ are statistically independent

τ can often be “eyeballed” by looking at history plots

Autocorrelation issues

- ① It is often crucial to make the autocorrelation time as short as possible
→ improved simulation algorithms may be needed
- ② Autocorrelations increase dramatically near phase transitions
→ critical slowing down
- ③ Statistical uncertainties are underestimated if autocorrelations are ignored
→ multiply errors by τ or use robust error estimation methods

Alternatives to Metropolis

Reminder: Metropolis acceptance probabilities

$$A_{XY} = \begin{cases} 1 & \text{if } H(Y) \leq H(X) \\ e^{-\beta\Delta H} & \text{if } H(Y) > H(X) \end{cases}$$

Detailed balance can be satisfied by other acceptance probabilities.

- 1 Glauber algorithm:

$$A_{XY} = \frac{1}{1 + e^{\beta\Delta H}}$$

- 2 Heat-bath algorithm
- 3 Overrelaxation

Ising model

Simple model of ferromagnetism

We have a lattice of spins $\sigma_i = \pm 1$

$$H = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j - B \sum_i \sigma_i; \quad \sum_{\langle ij \rangle} = \text{sum over nearest neighbours}$$

Phase transition

For $B = 0$ there is a second order phase transition at temperature T_c

$T > T_c$: Average magnetisation $\mu = 0$

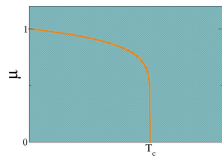
$T < T_c$: Spontaneous magnetisation $\mu \neq 0$

Critical behaviour: with $t \equiv (T_c - T)/T_c$,

$$\mu \sim t^\beta, \quad \chi \sim t^{-\gamma}, \quad C_v \sim (-t)^{-\alpha}$$

The **critical exponents** $\alpha, \beta, \gamma, \dots$ are **universal**:

Same behaviour for **all** systems with the same symmetries and dimensions



Simulation of Ising model

No analytical solution is known for $d = 3$ or $B \neq 0$

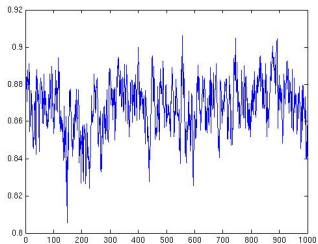
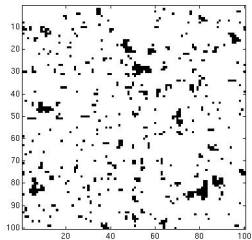
— Monte Carlo simulations are essential

Metropolis algorithm

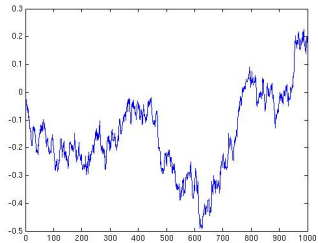
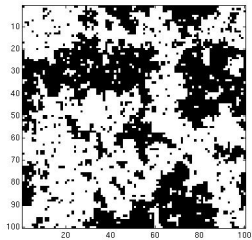
- ① Start with all $\sigma_i = 1$ (cold start) or random ± 1 (hot start)
- ② Sweep through all sites of lattice successively.
For each site:
 - ▶ Calculate the energy difference ΔE if you flip the spin σ_i
 - ▶ If $\Delta E < 0$ flip the spin
 - ▶ otherwise generate a uniform random $r \in (0, 1)$,
flip the spin if $r < e^{-\beta \Delta E}$
- ③ Compute physical quantities (μ , E etc)
- ④ Repeat for N sweeps

Simulation results

$$T \approx 0.9 T_c$$



$$T \approx T_c$$



$$T \approx 1.13 T_c$$

Finite volume issues

Boundary conditions

We want to study a macroscopic (nearly infinite) system

- the finite system we simulate is only a subsample of this
- use **periodic** boundary conditions to mimic surroundings

No phase transitions on a finite volume

All thermodynamic functions on finite V are smooth functions of T .

Not insurmountable: use **finite volume scaling**

- As V grows the crossover becomes sharper
- Extrapolate to $1/V = 0$
- There are critical exponents for finite volume scaling!

Summary

Markov processes

- A Markov process is a stochastic process with no memory
- Described by transition probabilities
- All Markov processes obey the **master equation**
- **Detailed balance** is a **sufficient** condition for a stationary process

Markov chain Monte Carlo

- Use an ergodic Markov chain to create a statistical distribution of configurations
- **Metropolis**: all-purpose algorithm for Monte Carlo simulations
- Straightforward application to condensed matter systems
eg **Ising model** → **lab!**
- Autocorrelations need to be monitored